Table 6.1

GANTT CHART

Task ID	Task Name	Task Lead	Start Date	End Date	JUN 2026				JULY 2026				AUG 2026				SEPT 2026				OCT 2026				NOV 2026			
					W 1	W 2	W 3	W4	W 1	W 2	W 3	W 4	W 1	W 2	W 3	W 4	W 1	W 2	W 3	W 4	W 1	W 2	W 3	W 4	W 1	W 2	W 3	W 4
1	**INTRODUCTION**																											
1.1	Create an introduction	Basubas	June 25	June 27																								
1.2	Explain the problem, research gap and solution	Isagunde	June 25	June 27																								
1.3	Write the specific objectives	Basubas	June 27	June 28																								
1.4	Define what the system includes and excludes	Isagunde	June 27	June 28																								
1.5	Explain who benefits from the study	Catarig	June 25	June 28																								
1.6	Show the study process	Catarig	June 28	June 29																								
1.7	Define the definitions of terms	Borces	June 25	June 29																								
1.8	Make the questionnaire	Basubas	June 25	July 1																								
1.9	Collect responses	Isagunde	June 25	July 1																								
1.10	Add citations	Basubas	June 30	July 1																								
1.11	Revise Chapter 1	All	June 30	July 1																								
2	**REVIEW OF RELATED LITERATURE AND STUDIES**																											
2.1	Identify and explain the theories	Catarig	July 2	July 4																								
2.2	Research journals	Borces	July 2	July 5																								
2.3	Gather and summarize literature	Isagunde	July 2	July 5																								
2.4	Review related studies	Basubas	July 2	July 5																								
2.5	Compare DriveAlert with existing systems and identify the research gap	Isagunde	July 2	July 6																								
2.6	Add citations, organize references, and revise Chapter 2	Basubas	July 6	July 8																								
3	**RESEARCH METHODOLOGY**																											
3.1	Select and explain the Agile methodology	Isagunde	July 4	July 5																								
3.2	Define the system requirements and development plan	Basubas	July 4	July 7																								
3.3	Prepare the Business Model Canvas	Basubas	July 4	July 8																								
3.4	Create the Driver and Trusted Contact program workflows	Borces	July 5	July 7																								
3.5	Complete the Validation Boards for Stages 1 and 2	Basubas	July 7	July 9																								
3.6	Prepare the Gantt charts	Isagunde	July 6	July 10																								
3.7	Create the Functional Decomposition Diagram	Catarig	July 4	July 6																								
3.8	Create the Use Case Diagram	Basubas	July 9	July 10																								

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3.9	Design the mobile and web user interfaces	Borces	July 5	July 7																								
3.10	Create the storyboard	Basubas	July 9	July 10																								
3.11	Identify the required database entities, attributes, and relationships	Borces	July 6	July 9																								
3.12	Create the Entity Relationship Diagram	Basubas	July 7	July 10																								
3.13	Prepare the Data Dictionary	Catarig	July 7	July 9																								
3.14	Create the Network Design	Isagunde	July 6	July 10																								

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3.15	Create the Network Model	Isagunde	July 6	July 10																								
3.16	Describe the system development	Basubas	July 10	July 11																								
3.17	Create the Technology Stack Diagram	Basubas	July 10	July 11																								
3.18	Define the software and hardware specifications	Borces	July 10	July 11																								
3.19	Prepare the Program Specification	Catarig	July 10	July 11																								
3.20	Prepare the Team Composition	Isagunde	July 10	July 10																								
3.21	Prepare the Appendices	Basubas	July 12	July 14																								
3.22	Prepare the List of Modules	Basubas	July 12	July 13																								
3.23	Revise the Gantt Chart and Agile Sprint Plan	Isagunde	July 13	July 14																								
3.24	Revise Chapter 3	All	July 12	July 15																								
3.25	Finalize the manuscript	All	July 16	July 22																								
4	**DEVELOPMENT TRANSITION PREPARATION**																											
4.1	Review the finalized manuscript requirements, diagrams, and module boundaries	All	July 18	July 20																								
4.2	Prepare development tools, accounts, hardware requirements, and project resources	Basubas and Borces	July 18	July 22																								
4.3	Assign module ownership, repository workflow, and coding standards	All	July 20	July 22																								
5	**PROJECT FOUNDATION, IOT ASSEMBLY, AND APPLICATION SETUP**																											
5.1	Create the Git repository and configure the Android, web, Firebase, and development environments	Basubas, Isagunde, and Borces	July 23	July 28																								
5.2	Finalize the local and cloud database schema and account structures	Borces	July 23	July 29																								
5.3	Acquire, inspect, and verify the compatibility of the IoT components	Basubas and All	July 23	July 27																								
5.4	Assemble the camera, infrared LEDs, onboard speaker, power connections, and mounting	Basubas with All	July 23	August 1																								
5.5	Configure the camera firmware, infrared operation, resolution, and frame rate	Basubas	July 29	August 3																								
5.6	Test camera capture, infrared illumination, speaker playback, wiring, and power stability	Basubas and Catarig	August 1	August 5																								
5.7	Develop the Android starter, registration, login, role selection, and Driver Setup screens	Basubas and Isagunde	July 24	August 5																								
5.8	Develop web registration, login, routing, and the Shared Drivers dashboard foundation	Isagunde and Catarig	July 24	August 5																								
5.9	Create authentication records and initial database access-control rules	Borces	July 27	August 5																								
6	**IOT COMMUNICATION AND CORE APPLICATION DEVELOPMENT**																											
6.1	Develop the local offline Wi-Fi connection with Android as host and the IoT device as client	Basubas	August 6	August 12																								
6.2	Implement the IoT HTTP video stream and Android video receiving	Basubas	August 8	August 16																								
6.3	Implement the Android-to-IoT warning command and onboard speaker activation	Basubas and Catarig	August 10	August 16																								
6.4	Test connection loss, reconnection, video-stream stability, and alarm communication	Basubas and Borces	August 14	August 19																								

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6.5	Develop Driver Home, Monitoring Session, device status, and monitoring-readiness functions	Basubas and Isagunde	August 6	August 19																								
6.6	Develop Alert History, Alert Detail, basic drowsiness insights, Settings, and warning-sound functions	Isagunde and Catarig	August 9	August 19																								
6.7	Develop the web Shared Drivers, shared records, alert summaries, synchronization status, and account views	Isagunde and Catarig	August 6	August 16																								
6.8	Connect the mobile and web account functions to the database and approved-driver access rules	Borces	August 8	August 19																								
7	**MEDIAPIPE FACE LANDMARKER INTEGRATION**																											
7.1	Integrate MediaPipe Face Landmarker into the Android video-processing flow	Basubas	August 20	August 25																								
7.2	Build the live frame-processing pipeline using the received IoT camera frames	Basubas and Borces	August 22	August 30																								
7.3	Identify and validate the required eye, mouth, face, and head-position landmarks	Borces	August 23	August 28																								
7.4	Calculate Eye Aspect Ratio, Mouth Aspect Ratio, and head-pitch measurements	Basubas	August 26	September 2																								
7.5	Implement face-visibility, camera-adjustment, and landmark-stability feedback	Isagunde	August 28	September 2																								
7.6	Optimize frame-processing speed, detection delay, and Android compatibility	Basubas	August 29	September 2																								
7.7	Test landmark extraction in daytime, low-light, and infrared-assisted conditions	All	August 30	September 2																								
8	**DROWSINESS-SIGN DETECTION, RECORDS, AND SYNCHRONIZATION**																											
8.1	Implement prolonged-eye-closure detection using Eye Aspect Ratio	Basubas	September 3	September 8																								
8.2	Implement yawning detection using Mouth Aspect Ratio	Catarig	September 3	September 8																								
8.3	Implement pitch-based head-nodding detection	Basubas	September 4	September 10																								
8.4	Implement duration, repetition, temporal-confirmation, and warning-decision rules	Basubas	September 7	September 12																								
8.5	Record controlled samples and calibrate EAR, MAR, pitch, duration, and repetition thresholds	All	September 8	September 16																								
8.6	Connect confirmed drowsiness-sign detections to the Monitoring Session and onboard speaker warning	Basubas and Isagunde	September 10	September 16																								
8.7	Implement monitoring-session and drowsiness-alert records	Borces	September 9	September 14																								
8.8	Implement Trusted Contact requests, approval status, Driver relationships, and access control	Borces and Catarig	September 9	September 16																								
8.9	Implement offline storage, synchronization queue, retry handling, status, and basic drowsiness insights	Basubas and Borces	September 11	September 16																								
9	**SYSTEM INTEGRATION**																											
9.1	Integrate mobile authentication, interfaces, and database operations	Isagunde and Borces	September 17	September 22																								
9.2	Integrate the IoT video stream with the Monitoring Session	Basubas	September 17	September 24																								
9.3	Integrate drowsiness-sign detection with the received live IoT camera frames	Basubas and Borces	September 19	September 26																								
9.4	Integrate threshold decisions with the warning command and onboard speaker	Basubas and Isagunde	September 22	September 28																								

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9.5	Integrate monitoring sessions, alerts, alert summaries, offline storage, and synchronization	Basubas, Borces, and Catarig	September 20	September 29																								
9.6	Integrate the Trusted Contact web portal with approved Driver records	Isagunde and Catarig	September 19	September 27																								
9.7	Resolve initial communication, database, alarm, access-control, and synchronization defects	All	September 27	September 30																								
10	**MOBILE AND WEB MODULE TESTING**																											
10.1	Test registration, login, role selection, account access, and Driver Setup	Borces and Catarig	October 1	October 4																								
10.2	Test the Monitoring Session, device status, camera feedback, and mobile navigation	Isagunde	October 1	October 6																								
10.3	Test Alert History, Alert Detail, basic drowsiness insights, Settings, and warning-sound functions	Catarig and Isagunde	October 3	October 8																								
10.4	Test Trusted Contact requests, approval, mobile dashboard, and shared records	Borces and Catarig	October 5	October 10																								
10.5	Test web registration, login, dashboards, records, alert summaries, and account views	Isagunde and Catarig	October 3	October 11																								
10.6	Verify approved-driver access control, security rules, validation, and duplicate prevention	Borces	October 7	October 12																								
10.7	Correct module defects and perform regression testing	All	October 10	October 14																								
11	**COMPLETE SYSTEM AND CONTROLLED TESTING**																											
11.1	Test the IoT camera, infrared illumination, speaker, wiring, and power stability	Basubas and Borces	October 15	October 18																								
11.2	Test offline Wi-Fi, video transmission, disconnection, reconnection, and stream stability	All	October 16	October 21																								
11.3	Test prolonged eye closure, yawning, and head nodding under controlled conditions	All	October 18	October 23																								
11.4	Evaluate thresholds, repeated detections, false alerts, missed detections, and response delay	Basubas and Borces	October 20	October 25																								
11.5	Test mobile-to-IoT warning commands and onboard speaker response	Isagunde and Basubas	October 19	October 24																								
11.6	Test synchronization and Trusted Contact record viewing with and without internet	Catarig and Borces	October 20	October 25																								
11.7	Conduct end-to-end functional testing and record test cases, results, and issues	All	October 23	October 28																								
11.8	Prepare the controlled pilot-evaluation protocol, forms, and system setup	All	October 25	October 28																								
12	**ISSUE RESOLUTION AND RELEASE CANDIDATE**																											
12.1	Consolidate, classify, and prioritize all identified defects	All	October 29	October 31																								
12.2	Correct mobile interface, navigation, validation, authentication, and account defects	Basubas and Isagunde	November 1	November 6																								
12.3	Correct web display, responsiveness, navigation, and access-control defects	Isagunde and Catarig	November 1	November 5																								
12.4	Correct IoT connection, camera, infrared, streaming, speaker, and power defects	Basubas and Borces	November 1	November 7																								
12.5	Correct drowsiness-sign detection, threshold, false-alert, and response-delay defects	Basubas and Borces	November 1	November 8																								
12.6	Correct database, records, synchronization, retry, and security-rule defects	Borces and Catarig	November 3	November 8																								

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12.7	Perform complete regression testing and prepare the release-candidate build	All	November 9	November 11																								
12.8	Update source-code organization, comments, diagrams, and technical documentation	Catarig and Isagunde	November 7	November 11																								
13	**DEPLOYMENT, FINAL EVALUATION, AND DOCUMENTATION**																											
13.1	Prepare and verify the Android application build	Basubas and Isagunde	November 12	November 16																								
13.2	Deploy the Trusted Contact web portal, cloud database, and security rules	Isagunde and Borces	November 12	November 16																								
13.3	Upload the final IoT firmware and configure the camera, IR LEDs, speaker, and enclosure	Basubas and Borces	November 12	November 17																								
13.4	Install and configure the system on selected controlled-test devices	All	November 15	November 17																								
13.5	Conduct the final controlled system and pilot evaluation	All	November 18	November 20																								
13.6	Analyze the evaluation results and review the system against the objectives	Basubas and Catarig	November 20	November 22																								
13.7	Update the manuscript, ERD, Data Dictionary, workflows, diagrams, specifications, and appendices	Isagunde and Borces	November 20	November 24																								
13.8	Prepare the final presentation, demonstration flow, and supporting materials	All	November 21	November 25																								

LEGEND	COMPLETE
	ONGOING
	INCOMPLETE

This section displays tasks and their associated timeframes in a horizontal bar format, enabling researchers to organize, track, and coordinate activities efficiently. It provides a comprehensive overview of the project's progress, improves communication, and ensures that project requirements are completed on schedule. Using the Agile Methodology, the Gantt chart is helpful due to the methodology's iterative nature, allowing each phase to be clearly represented and tracked